

PHARMACOLOGY ASSIGNMENT

Topic: Antihistamines — Salt Forms, Salt Base & Doses

Introduction

Pharmaceutical salts are ionic forms of drug molecules formed by reacting the parent drug (base or acid) with a suitable counter-ion. Salt formation is one of the most widely used strategies in drug development to improve solubility, stability, bioavailability, and palatability. Antihistamines — drugs that competitively block histamine H₁ receptors — are available in numerous salt forms depending on the generation and clinical use. This assignment outlines the parent drug, its available salt forms, the corresponding salt base (counter-ion), and the standard therapeutic dose for each major antihistamine.

Salt Forms, Salt Base & Dose of Antihistamines

Parent Drug	Salt / Ester Form	Salt Base (Counter-ion)	Standard Dose
FIRST-GENERATION ANTIHISTAMINES			
<i>Diphenhydramine</i>	Diphenhydramine Hydrochloride Diphenhydramine Citrate Diphenhydramine Tannate	HCl (Hydrochloride) Citric acid Tannic acid	25–50 mg every 4–6 hrs 38 mg every 4–6 hrs 25–50 mg every 6 hrs
<i>Chlorpheniramine</i>	Chlorpheniramine Maleate Chlorpheniramine Tannate	Maleic acid Tannic acid	4 mg every 4–6 hrs 8–12 mg every 12 hrs
<i>Promethazine</i>	Promethazine Hydrochloride Promethazine Theoclate	HCl (Hydrochloride) Theophylline-8-ox yacetic acid	12.5–25 mg every 4–6 hrs 25 mg at night
<i>Hydroxyzine</i>	Hydroxyzine Hydrochloride Hydroxyzine Pamoate	HCl (Hydrochloride) Pamoic acid (Embonate)	25–100 mg/day in divided doses 25–100 mg/day in divided doses
<i>Cyproheptadine</i>	Cyproheptadine Hydrochloride	HCl (Hydrochloride)	4 mg 3–4 times daily (max 32 mg/day)

Parent Drug	Salt / Ester Form	Salt Base (Counter-ion)	Standard Dose
<i>Brompheniramine</i>	Brompheniramine Maleate Brompheniramine Tannate	Maleic acid Tannic acid	4 mg every 4–6 hrs 6–12 mg every 12 hrs
<i>Clemastine</i>	Clemastine Fumarate	Fumaric acid	1.34 mg twice daily (max 8.04 mg/day)
<i>Triprolidine</i>	Triprolidine Hydrochloride	HCl (Hydrochloride)	2.5 mg every 4–6 hrs
<i>Doxylamine</i>	Doxylamine Succinate	Succinic acid	25 mg at bedtime
SECOND-GENERATION ANTIHISTAMINES			
<i>Cetirizine</i>	Cetirizine Hydrochloride Cetirizine Dihydrochloride	HCl (Hydrochloride) Di-HCl	10 mg once daily 10 mg once daily
<i>Levocetirizine</i>	Levocetirizine Dihydrochloride	Di-HCl (Dihydrochloride)	5 mg once daily (evening)
<i>Loratadine</i>	Loratadine (free base) Loratadine Hydrochloride	Free base (no salt) HCl (Hydrochloride)	10 mg once daily 10 mg once daily
<i>Desloratadine</i>	Desloratadine (free base)	Free base (no salt)	5 mg once daily
<i>Fexofenadine</i>	Fexofenadine Hydrochloride	HCl (Hydrochloride)	60 mg twice daily OR 180 mg once daily
<i>Azelastine</i>	Azelastine Hydrochloride	HCl (Hydrochloride)	Nasal: 1–2 sprays/nostril twice daily Ophthalmic: 1 drop twice daily
<i>Bilastine</i>	Bilastine (free base)	Free base (no salt)	20 mg once daily (on empty stomach)
<i>Rupatadine</i>	Rupatadine Fumarate	Fumaric acid	10 mg once daily
<i>Ebastine</i>	Ebastine (free base)	Free base (no salt)	10–20 mg once daily
<i>Mizolastine</i>	Mizolastine (free base)	Free base (no salt)	10 mg once daily

Parent Drug	Salt / Ester Form	Salt Base (Counter-ion)	Standard Dose
Olopatadine	Olopatadine Hydrochloride	HCl (Hydrochloride)	Ophthalmic: 1–2 drops twice daily Nasal: 2 sprays/nostril twice daily

Important Notes

- Salt vs. Free Base:** Most antihistamines are weak bases marketed as acid-addition salts (e.g., hydrochlorides, maleates) to improve water solubility. Some newer second-generation agents (loratadine, desloratadine, bilastine) are used as free bases.
- Counter-ion Choice:** The counter-ion (acid used to form the salt) influences solubility, stability, and taste. Hydrochloride salts are most common; maleate and fumarate salts improve stability; tannate salts provide slow release (sustained action).
- Dose Equivalence:** Doses listed refer to the salt form as commercially available. The free-base equivalent weight differs due to the molecular weight of the counter-ion.
- Tannate Formulations:** Diphenhydramine tannate and chlorpheniramine tannate are used in pediatric suspensions to reduce bitterness and provide prolonged action.
- Pamoate Salts:** Hydroxyzine pamoate has reduced systemic absorption compared to HCl salt, but equivalent therapeutic activity at the same dose.
- Routes of Administration:** Doses listed are primarily for oral administration unless otherwise specified (e.g., azelastine — nasal/ophthalmic; promethazine — also IV/IM).

Conclusion

The choice of salt form in antihistamine formulation is a critical aspect of pharmaceutical development. Hydrochloride and maleate salts dominate due to their superior aqueous solubility and chemical stability. Tannate salts serve specialized roles in pediatric and sustained-release preparations. Understanding the relationship between parent drug, salt form, counter-ion, and therapeutic dose is essential for pharmacists and clinicians to ensure appropriate drug selection, dosing accuracy, and formulation substitution in clinical practice.